

**Do not hand in this or any of the following question sheets.**  
**Nothing on them will be graded.**

**Problem 1:**

(4)

*These are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.*

**Question 1:** Consider the Two-Phase-Commit Protocol in case of a transaction  $T$  with 1 coordinator and 3 subordinates. In which of the following scenarios will transaction  $T$  eventually commit?

- (a) Suppose that the coordinator force-writes a commit log but crashes before it can send a commit message to the subordinates.
- (b) Suppose that the coordinator sends a commit message to all subordinates, two of the subordinates receive the commit message and force-write a commit log but one of the subordinates crashes before receiving the commit message.
- (c) Suppose that the coordinator sends a commit message to all subordinates, all subordinates receive the commit message but only one of them manages to force-write a commit log; the other two subordinates crash immediately after receiving the commit message.
- (d) Suppose that the coordinator sends a commit message to all subordinates, all subordinates receive the commit message and force-write a commit log; but only one of the subordinates manages to send an ACK message to the coordinator; the other two subordinates crash before they manage to send an ACK message.

**(Correct: a,b,c,d)**

*Explanation:*

Answers a, b, c, and d are correct. See the Slides III.1, p.40.

**Question 2:** Which of the following statements concerning scalar or vector clocks are correct? In all these statements,  $C$  denotes the global clock, “ $\rightarrow$ ” denotes the “happened-before” relation, and  $e, e'$  denote two distinct events.

- (a) For a scalar clock  $C$ : if  $C(e) < C(e')$ , then it is guaranteed that  $e \rightarrow e'$  holds.
- (b) For a scalar clock  $C$ : if  $e \rightarrow e'$ , then it is guaranteed that  $C(e) < C(e')$  holds.
- (c) For a vector clock  $C$ : if  $C(e) < C(e')$ , then it is guaranteed that  $e \rightarrow e'$  holds.
- (d) For a vector clock  $C$ : if  $e \rightarrow e'$ , then it is guaranteed that  $C(e) < C(e')$  holds.

**(Correct: b,c,d)**

*Explanation:*

Answers b, c, and d are correct. See the Slides III.2, p.46.

**Question 3:** Which of the following statements concerning the CAP Theorem and its extension to PACELC are correct?

- (a) The following is a consequence of the CAP Theorem: if a distributed data store guarantees partition tolerance and availability then it cannot guarantee consistency.
- (b) The following is a consequence of the CAP Theorem: if a distributed data store guarantees partition tolerance and consistency then it cannot guarantee availability.
- (c) “Availability” in the CAP Theorem refers to the requirement that, in a distributed data store, at least one node always has to be up and running.
- (d) In PACELC, the first “C” stands for consistency and the second “C” stands for “correctness”.

**(Correct: a,b)**

*Explanation:*

Answers a and b are correct: see Slides III.2, p.38. For answer c, see p.37. For answer d, see p.40.

**Question 4:** Which of the following read/write quorums are allowed in the Quorum-Consensus protocol?

- (a) In case of replication factor  $N = 17$ :  $w = 8$  and  $r = 8$ .
- (b) In case of replication factor  $N = 17$ :  $w = 8$  and  $r = 9$ .
- (c) ROWA (read one, write all):  $w = N$  and  $r = 1$ , where  $N$  denotes the replication factor.
- (d) RAWO (read all, write one):  $r = N$  and  $w = 1$ , where  $N$  denotes the replication factor.

**(Correct: c)**

*Explanation:*

Answer c is correct. See Slides III.2, p.41.

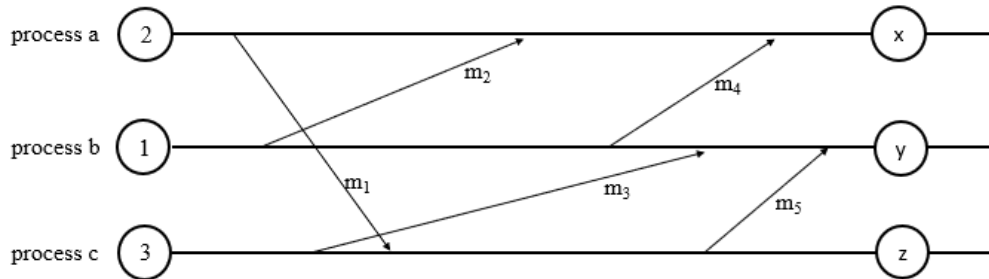
## Problem 2:

(4)

Below are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.

### Scalar Clocks: [1 credit]

Consider a scenario with 3 processes. We observe these processes in a small time window. The processes have already started. Therefore, at the beginning of the time window, the clock counts are not 0. Instead they are 2, 1, and 3, respectively. Within this time window, 5 messages  $m_1, \dots, m_5$  are sent between processes. Each of the circles labeled  $x, y, z$ , respectively, stand for a particular point in time. Your task is to determine the values for the clocks at each of these points in time.



**Question 5:** Which of the following clock values are correct:

- (a)  $x = 4$       (b)  $y = 7$       (c)  $z = 4$       **(Correct: b)**

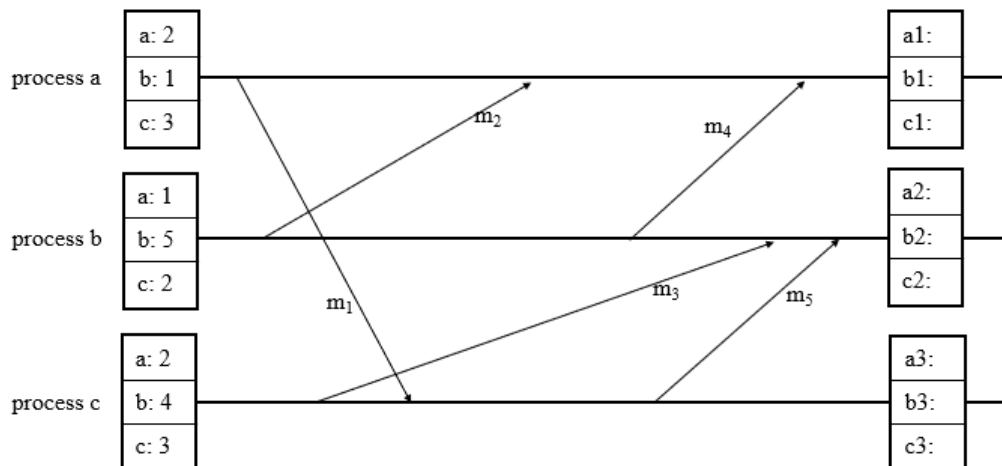
*Explanation:*

Answer b is correct. See the Slides III.2, p.44 and the sample solution in TUWEL.

### Vector Clocks: [3 credits]

Consider a scenario with 3 processes. We observe these processes in a small time window. The processes have already started. Therefore, at the beginning of the time window, the vector clocks have values different from  $(0,0,0)$ . Instead, they are  $(2,1,3)$ ,  $(1,5,2)$ , and  $(2,4,3)$ , respectively. Within this time window, 5 messages  $m_1, \dots, m_5$  are sent between processes.

At the end of the window, we want to know the final vector clock values at each process, represented by the labels  $(a1, b1, c1)$ ,  $(a2, b2, c2)$ , and  $(a3, b3, c3)$ , respectively. Your task is to determine the correct values of these clocks at the end of the window, based on vector clock rules.



**Question 6:** Which of the following clock values are correct for process a:

- (a)  $a1 = 4$       (b)  $b1 = 7$       (c)  $c1 = 3$       **(Correct: b,c)**

*Explanation:*

Answers b and c are correct. See the Slides III.2, p.47 and the sample solution in TUWEL.

**Question 7:** Which of the following clock values are correct for process b:

(a)  $a2 = 3$       (b)  $b2 = 8$       (c)  $c2 = 5$       (Correct: a)

*Explanation:*

Answer a is correct. See the Slides III.2, p.47 and the sample solution in TUWEL.

**Question 8:** Which of the following clock values are correct for process  $c$ :

(a)  $a3 = 3$       (b)  $b3 = 4$       (c)  $c3 = 5$       (Correct: a,b)

*Explanation:*

Answers a and b are correct. See the Slides III.2, p.47 and the sample solution in TUWEL.

### Problem 3:

(3)

*Below are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.*

Consider a relational database with information on students and lectures as well as information as to which student has already taken the exam of which lecture (together with the date of the exam). Suppose that the (very simplified) schema is as follows:

Students (s\_id, name, studies),  
Exams (student, lecture, date) and  
Lectures (l\_id, title, prof)

where Exams.student and Exams.lecture are foreign keys into the Students and Lectures table, respectively. Suppose that these tables have the following content:

| Students |       |         | Exams   |         |      | Lectures |       |       |
|----------|-------|---------|---------|---------|------|----------|-------|-------|
| s_id     | name  | studies | student | lecture | date | l_id     | title | prof  |
| S1       | alice | 645     | S1      | L1      | 2024 | L1       | ADBS  | carol |
| S2       | bob   | 937     | S2      | L1      | 2025 | L2       | ThInf | david |
|          |       |         | S1      | L2      | 2023 | L3       | EP2   | emma  |

Below you find three realizations (with gaps) of this contents as databases in a document store.

### Database DB1

```
Exams: [
  { "student": "S1",
    "name": "alice",
    "studies": 645,
    

Q9a

 },
  { "student": "S2",
    "name": "bob",
    "studies": 937,
    

Q9b

 },
  { 

Q9c


    "lecture": "L2",
    "title": "ThInf",
    "prof": "david",
    "date": 2023
  }
]
```

### Database DB2

```
Lectures: [
  { "lId": "L1",
    "title": "ADBS",
    "prof": carol,
    "exams":
      [ { "student": "S1",
          "name": "alice",
          "studies": 645,
          "date": 2024 }
        ],
    

Q10a


  },
  { "lId": "L2",
    "title": "ThInf",
    "prof": david,
    "exams":
      [ 

Q10b


        ],
    },
  { "lId": "L3",
    "title": "EP2",
    "prof": "emma",
    "exams":
      

Q10c


    }
]
```

### Database DB3

```
Students:
[ { "s_id": "S1",
  "name": "alice",
  "studies": 645,
  "exams": ["L1","L2"] },
  { "s_id": "S2",
    "name": "bob",
    "studies": 937,
    

Q11a


  } ]

Lectures:
[ { "lId": "L1",
  "title": "ADBS",
  "prof": "carol",
  

Q11b


},
  { "lId": "L2",
    "title": "ThInf",
    "prof": "david" ,
    

Q11c


  },
  { "lId": "L3",
    "title": "EP2",
    "prof": "emma",
    

Q11d


  } ]
```

**Question 9:** Suppose that **Database DB1** has been designed in such a way that the entire information on Students, Exams, and Lectures is contained in a single collection. More specifically, the Students and Lectures collections are denormalized into Exams. Which of the following statements are correct if your task is to fill in the missing parts into Database DB1. In particular, you should replace the boxes Q9a, Q9b, and Q9c by the correct code snippets.

- (a)  has to be replaced by the following code snippet:  
`"lecture": "L1", "title": "ADBS", "prof": "carol", "date": 2024`
- (b)  has to be replaced by the following code snippet:  
`"lecture": "L1", "title": "ADBS", "prof": "carol", "date": 2024`
- (c)  has to be replaced by the following code snippet:  
`"student": "S2", "name": "bob", "studies": 937,`
- (d) The type of denormalization applied in Database DB1 is called “denormalizing into the one-side”.

(Correct: a)

*Explanation:*

Answer a is correct. See Slides III.3, p.24/25 and the sample solution in TUWEL.

**Question 10:** Suppose that **Database DB2** has been designed in such a way that the Lectures collection contains the entire information on Students having taken that Exam. Which of the following statements are correct if your task is to fill in the missing parts into Database DB2. In particular, you should replace the boxes Q10a, Q10b, and Q10c by the correct code snippets.

- (a)  has to be replaced by the following code snippet:  
`{ "student": "S2", "name": "bob", "studies": 937, "date": 2025 }`
- (b)  has to be replaced by the following code snippet:  
`{ "student": "S2", "name": "bob", "studies": 937, "date": 2025 }`
- (c)  has to be left empty (i.e., there is nothing missing here)
- (d) One motivation for choosing this type of denormalization could be to speed up the following type of queries: for a given lecture, obtain detailed information on students having taken this exam.

(Correct: a,d)

*Explanation:*

Answers a,d are correct. See Slides III.3, p.24/25 and the sample solution in TUWEL.

**Question 11:** Suppose that **Database DB3** has been designed in such a way that the Students collection contains for each student the list of l\_id's of the lectures whose exams she/he has already taken. Likewise, the Lectures collection contains for each lecture the list of s\_id's of the students that have already taken that exam. Due to lack of space, we are not showing the Exams collection, which is of course also contained in Database DB3. Which of the following statements are correct if your task is to fill in the missing parts into Database DB3. In particular, you should replace the boxes Q11a, Q11b, Q11c, and Q11d by the correct code snippets.

- (a)  has to be replaced by the following code snippet: `"exams": ["L1"]`
- (b)  has to be replaced by the following code snippet: `"exams": ["S1"]`
- (c)  has to be replaced by the following code snippet: `"exams": ["L2"]`
- (d)  has to be replaced by the following code snippet: `"exams": [ ]`

(Correct: a)

*Explanation:*

Answer a is correct. See Slides III.3, p.24/25 and the sample solution in TUWEL.

## Problem 4:

(4)

The questions below have exactly 1 correct answer. In the answer options a – e, you are presented 5 different values. Answer option f means, that none of the values presented in a – e is correct. The number of credits is given for each question separately.

Assume a distributed database with 3 sites and the relations **Customer** and **Order**. See the following tables for details of the scenario. You can assume that the relations are *non-fragmented* and all records (and attributes) are *fixed-size*. Note that the attributes shown are a subset of all attributes.

| Relation | Site | # Records | Record Size (byte) |
|----------|------|-----------|--------------------|
| Order    | 1    | 10 000    | 2000               |
| Customer | 2    | 800       | 1000               |

| Relation | Attribute | Size |
|----------|-----------|------|
| Order    | oid       | 10   |
| Order    | customer  | 10   |
| Order    | product   | 100  |
| Customer | cid       | 10   |
| Customer | address   | 200  |

Consider the following query:  $\pi_{product, address}(Order \bowtie_{customer=cid} Customer)$

In this problem, you are asked to compute the communication cost (number of bytes transferred) for the given query and three different evaluation strategies. In all cases, to achieve the minimal communication cost, **project as early as possible to the attributes needed to evaluate the query**. **Order.customer** is a foreign key to **Customer.cid**, hence *the join has a selectivity of  $\frac{1}{800}$* . To keep things simple, we write MB and kB as shorthand for  $10^6$  and  $10^3$  bytes, respectively.

**Question 12: [1 credit]** What is the communication cost if the query is evaluated at site 3?

- (a) 1.246 MB      (b) 1.268 MB      (c) 1.282 MB      (d) 2.188 MB      (e) 2.266 MB      (f) none      **(Correct: b)**

*Explanation:*

Answer b is correct: We only need to send the join attributes + product + address to site 3, i.e., from the **Order** relation, we have to send  $10000 * (10 + 100) = 1.1$  MB  
from the **Customer** relation, we have to send  $800 * (10 + 200) = 168$  kB  
 $\Rightarrow total = 1.268$  MB

**Question 13: [1 credit]** What is the communication cost if the query is evaluated at site 1 and the result is transferred to site 3?

- (a) 3.146 MB      (b) 3.168 MB      (c) 3.182 MB      (d) 4.088 MB      (e) 4.144 MB      (f) none      **(Correct: b)**

*Explanation:*

Answer b is correct:  
from the **Customer** relation, we have to send  $800 * (10 + 200) = 168$  kB as before.  
Due to the foreign key relationship, the join has 10,000 tuples, each of size  $(100 + 200) = 300$  bytes, i.e., 3 MB  
 $\Rightarrow total = 3.168$  MB

**Question 14: [2 credits]** Now suppose that the DDBMS first determines by a semi-join at site 2 those tuples of the **Customer** relation that have a join partner. For this semi-join, you may assume selectivity = 0.5 (i.e., half of the tuples in the **Customer** relation have a join partner; it also means that the **Order** relation has only 400 distinct values in the customer attribute).

Here are the steps that the DDBMS carries out: it first sends the relevant information of the **Order** relation to site 2 (assume duplicate elimination!) and determines by a semi-join with the **Customer** relation those tuples of the **Customer** relation that have a join partner. Then the DDBMS sends the relevant information of the **Customer** relation to site 1 and computes the join result. Finally, the join result is transferred from site 1 to site 3. What is the total communication cost for this query plan?

- (a) 2.146 MB      (b) 2.168 MB      (c) 2.182 MB      (d) 3.088 MB      (e) 3.144 MB      (f) none      **(Correct: b)**

rect: d)

*Explanation:*

Answer d is correct:

transfer  $\pi_{customer}(Order)$  to site 2:  $10 * 400 = 4000 = 4$  kB

(notice: 400 rather than 800 tuples due to duplicate elimination)

transfer the result of the semi-join to site 1:  $210 * 400 = 84000 = 84$  kB

then add the size of the join result: 3 MB

$\Rightarrow total = 3.088$  MB

## Problem 5:

(5)

*These are true multiple choice question. Any number of answers (between 0 and all) could be correct. 1 credit for each question.*

(5.1) Evaluate the following Cypher query on the Database given on the last sheet of the exam:

```
MATCH (c:City)-[:HAS_AIRPORT]->(:Airport)-[:TRAIN]->(:Airport)-[f:FLIGHT]-(a:Airport)
WHERE c.name = 'London' AND f.duration > 70
RETURN a.code
```

**Question 15:** Mark exactly those values that are returned by the query.

(a) TXL      (b) JFK      (c) HND      (d) NRT      (e) CDG      (f) LGW

(Correct: a, c,e)

(5.2) Evaluate the following Cypher query on the Database given on the last sheet of the exam:

```
MATCH (:City {name: 'Tokyo'})-[:HAS_AIRPORT]->(:Airport)-[:FLIGHT|TRAIN*1..2]->(a2:Airport),
      (c2:City)-[:HAS_AIRPORT]->(a2)
RETURN DISTINCT c2.name
```

**Note:** :FLIGHT|TRAIN\*1..2 in Cypher is read as (:FLIGHT or :TRAIN)\*1..2

**Question 16:** Mark exactly those values that are returned by the query.

(a) London      (b) Paris      (c) Berlin      (d) Tokyo      (e) New York      (f) *The query is not valid*

(Correct: a,c,d)

(5.3) Evaluate the following Cypher query on the Database given on the last sheet of the exam:

```
MATCH p = allshortestPaths(
  (a1:Airport)-[:FLIGHT*]->(a2:Airport {code: 'TXL'})
)
WHERE a1 <> a2
return a1.code, length(p)
```

**Question 17:** Mark exactly those values that are returned by the query.

(a) LHR, 2      (b) HND, 2      (c) JFK, 4      (d) JFK, 3      (e) CDG, 1  
(f) TXL, 0      (g) LGW, 2      (h) CDG, 2      (i) LGW, 1      (j) HND, 3

(Correct: a,b,e,i)

(5.4) Evaluate the following Cypher query on the Database given on the last sheet of the exam:

```
MATCH (c:City)-[:HAS_AIRPORT]->(a:Airport)-[f:FLIGHT]->()
WHERE f.duration > 400
RETURN DISTINCT c.name, a.code
```

**Question 18:** Mark exactly those values that are returned by the query.

(a) London, LGW      (b) New York, JFK      (c) Berlin, TXL      (d) Paris, CDG  
(e) Tokyo, NRT      (f) London, LHR      (g) Tokyo, HND      (h) *The query is not valid*

(Correct: b,f,g)

(5.5) Which of the following Cypher queries correctly answers the question below?

*“List the cities that have at least one airport from which there is a direct train connection to another airport.”*

**Question 19:** Mark the queries that return the correct result.

- (a) `MATCH (c:City)-[:HAS_AIRPORT]->(a:Airport)`  
`MATCH (a)-[:FLIGHT]->(b:Airport)`  
`RETURN DISTINCT c.name`
- (b) `MATCH (a1:Airport)-[:TRAIN]->(a2:Airport)`  
`RETURN DISTINCT a2.code`
- (c) `MATCH (c:City)-[:HAS_AIRPORT]->(a:Airport)-[:TRAIN]->(b:Airport)`  
`RETURN DISTINCT c.name`
- (d) `MATCH (c:City)-[:HAS_AIRPORT]->(a:Airport)`  
`WHERE size((a)-[:TRAIN]->(b)) = 0`  
`RETURN DISTINCT c.name`

(Correct: c)

Overall: 20 points

## Graph DB Data Model

The graph contains nodes for cities with the label `:City` and nodes for airports with the label `:Airport`.

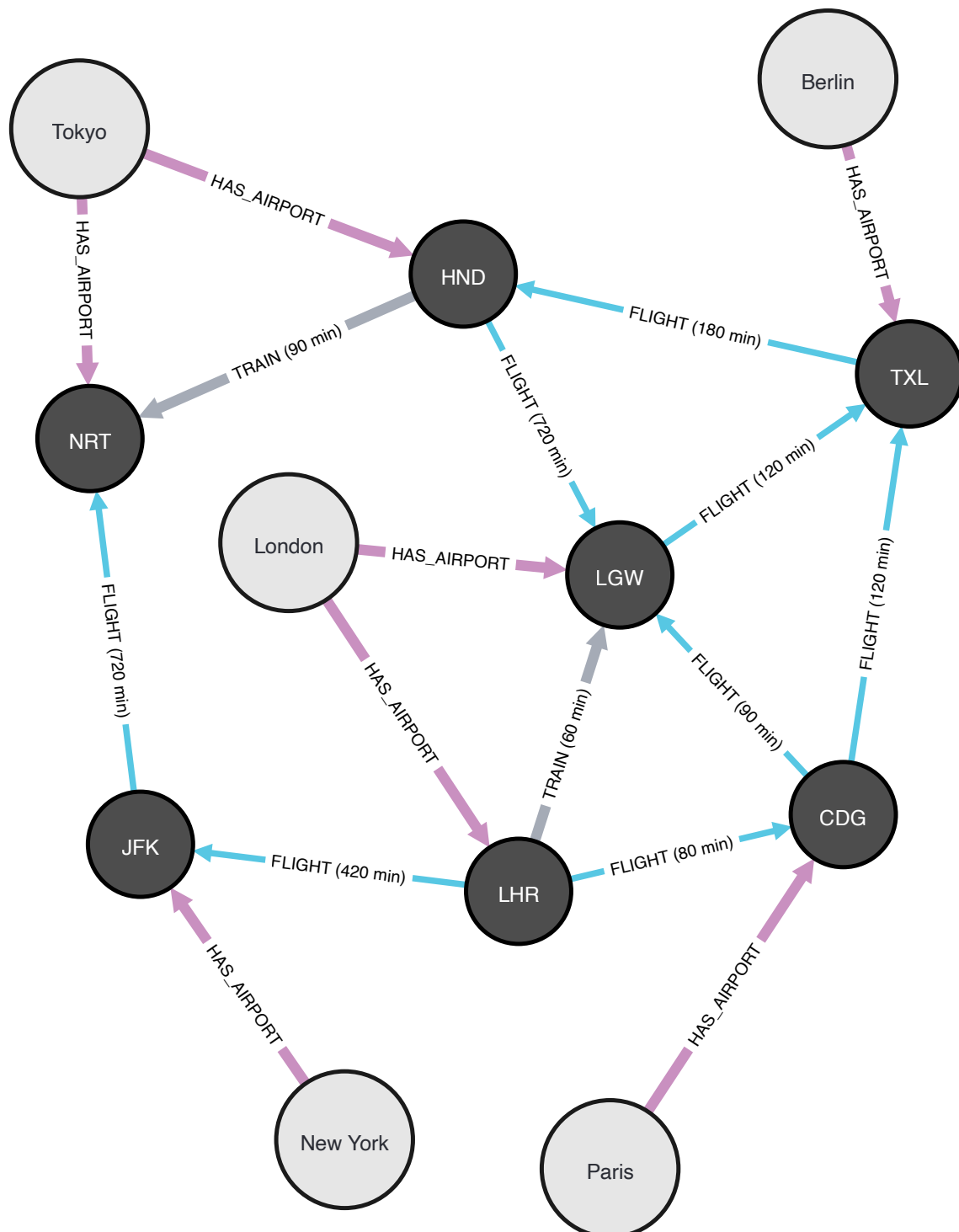
A city can be connected to one or more airports using a relationship of type `:HAS_AIRPORT`.

A relationship of type `:FLIGHT` between two airports represents a direct flight. Each `:FLIGHT` relationship has the property `duration`, which is an integer representing the duration of the flight in minutes.

A relationship of type `:TRAIN` between two airports represents a connection by train. Each `:TRAIN` relationship also has a `duration` property, containing the duration of the train journey in minutes.

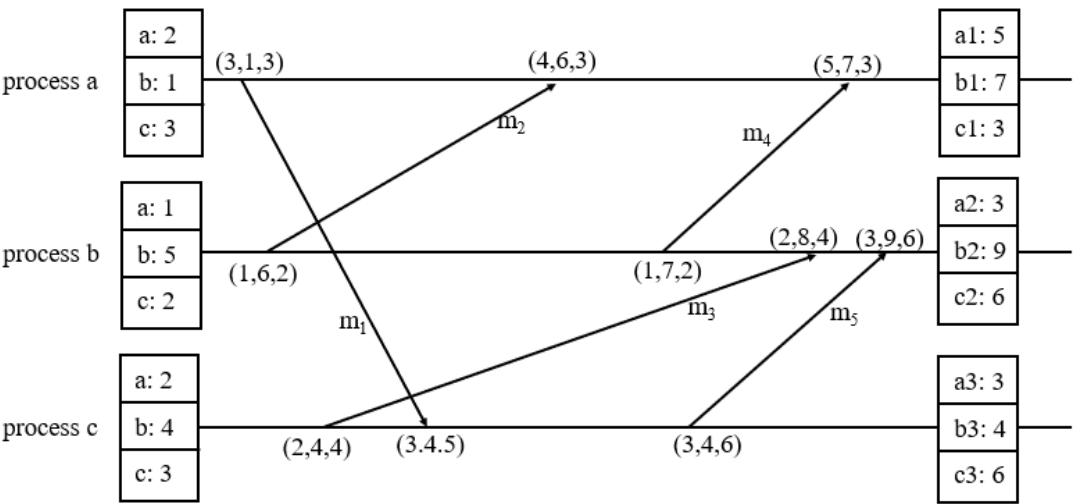
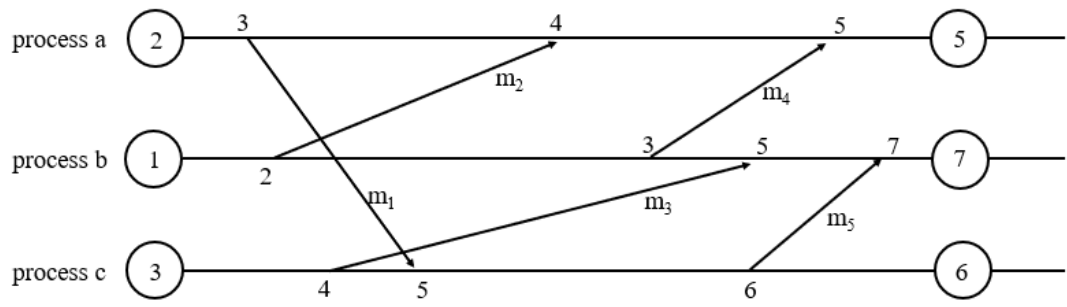
`:FLIGHT` and `:TRAIN` relationships cannot connect a node to itself.

## Graph Database



Work sheet for notes. Do not hand in. Will not be graded.

Sample Solution of Problem 2



### Sample Solution of Problem 3

#### Database DB1

```
Exams: [
{ "student": "S1",
  "name": "alice",
  "studies": 645,
  "lecture": "L1",
  "title": "ADBS",
  "prof": "carol",
  "date": 2024 },

{ "student": "S2",
  "name": "bob",
  "studies": 937,
  "lecture": "L1",
  "title": "ADBS",
  "prof": "carol",
  "date": 2025 },

{ "student": "S1",
  "name": "alice",
  "studies": 645,
  "lecture": "L2",
  "title": "ThInf",
  "prof": "david",
  "date": 2023 } ]
```

#### Database DB2

```
Lectures: [
{ "lid": "L1",
  "title": "ADBS",
  "prof": "carol",
  "exams":
  [ { "student": "S1",
      "name": "alice",
      "studies": 645,
      "date": 2024 },
    { "student": "S2",
      "name": "bob",
      "studies": 937,
      "date": 2025 } ]
},
{ "lid": "L2",
  "title": "ThInf",
  "prof": "david",
  "exams":
  [ { "student": "S1",
      "name": "alice",
      "studies": 645,
      "date": 2024 } ]
},
{ "lid": "L3",
  "title": "EP2",
  "prof": "emma",
  "exams":
  [ ]
} ]
```

#### Database DB3

```
Students:
[ { "s_id": "S1",
  "name": "alice",
  "studies": 645,
  "exams": ["L1","L2"] },

{ "s_id": "S2",
  "name": "bob",
  "studies": 937,
  "exams": ["L1"] } ]

Lectures:
[ { "lid": "L1",
  "title": "ADBS",
  "prof": "carol",
  "exams": ["S1","S2"] },

{ "lid": "L2",
  "title": "ThInf",
  "prof": "david",
  "exams": ["S1"] },

{ "lid": "L3",
  "title": "EP2",
  "prof": "emma",
  "exams": [ ] } ]

Exams:
[ { "student": "S1",
  "lecture": "L1",
  "date": 2024 },

{ "student": "S2",
  "lecture": "L1",
  "date": 2025 },

{ "student": "S1",
  "lecture": "L2",
  "date": 2023 } ]
```